

The Generalized Haldane (GH) model tracking population size changes and resolving paradoxes of genetic drift

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This study presents a **useful** model of genetic drift by incorporating variance in reproductive success, aiming to address several apparent paradoxes in molecular evolution. However, some of the apparent paradoxes only arise in the most basic version of standard models and have been reconciled in more advanced models. Nonetheless, this paper offers intuitive explanations for these apparent paradoxes, by adopting a new perspective and **solid** modeling and analysis. More broadly, the proposed model provides an alternative framework to address puzzling observations in molecular evolution, which will be of interest to evolutionary and population geneticists.

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Abstract

Population genetic models, such as the Wright-Fisher (WF) model, track relative gene frequencies. The absolute gene copy number, or population size (N), is supplied externally for tracking genetic drift. JBS Haldane (1927) proposed an alternative model based on the branching process, whereby each gene copy is transmitted to K descendants with the mean and variance of $E(K)$ and $V(K)$. In this model, $E(K)$ governs N , while $V(K)/N$ governs genetic drift. Nevertheless, as the branching process allows N to drift unboundedly, a Generalized Haldane (GH) model that regulates N more tightly is proposed. The GH model can account for several paradoxes of molecular evolution. Notably, genetic drift may often become stronger as N becomes larger in the ecological setting, thus contradicting the general view. In particular, a very small population growing exponentially experiences little drift. Interestingly, when the population grows and N oscillates near the carrying capacity, the paradoxical trend is also observed in both field works and laboratory experiments. This paradox whereby population size in genetics (N_e) and ecology (N) could be negatively correlated is resolved by the GH model. Additional paradoxes include ii) The two sexes experiencing drift differently; iii) Genetic drift of advantageous mutations being independent

of N ; iv) Multi-copy gene systems (viruses, mitochondria, etc.) having no definable N_e (for effective N). In brief, the GH model defines genetic drift simply as $V(K)$, or $V(K)/N$ averaged over the population. It represents an attempt at integrating genetical and ecological analyses into one framework.

Introduction

Genetic drift is defined as the random changes in frequency of genetic variants (Crow and Kimura 1970; Hartl and Clark 1997; Lynch, et al. 2016). Much, or even most, of DNA sequence evolution is governed by the random drift of neutral mutations. Nevertheless, even non-neutral variants are subjected to genetic drift as advantageous mutations often have a small probability of being fixed. If the strength of genetic drift is under-estimated, random changes that are missed in the analysis would result in the over-estimation of other evolutionary forces of greater biological interest, including selection, mutation, migration, meiotic drive and so on. In particular, the conclusion of pervasive selection at the molecular level has been suspected to be due to the failure to fully account for genetic drift (Crow and Kimura 1970; Fu 1997; Li 1997; Charlesworth 2009; Lynch, et al. 2016; Chen, Yang, et al. 2022).

Hagedoorn and Hagedoorn (1921) may be the first to suggest that random forces (accidents, for example) could impact long-term evolution. In the same decade, models of genetic drift were developed along two lines. The one approach that became widely adopted is the Wright-Fisher (WF) model (Crow and Kimura 1970; Hartl and Clark 1997), formalized by Fisher (1930) and Wright (1931). They define genetic drift to be due to the random sampling of genes. Thus, in generation transition, the variance in the frequency of a variant allele, x , would be

$$V(x) = \frac{x(1-x)}{f(N)} = \frac{x(1-x)}{N_e} \quad (1)$$

where N is the population size. (Here, we present the haploid model while a factor of 2 should be added to the diploid models.) The WF model has been modified in many directions (Wright 1969; Gillespie 1975; Eldon and Wakeley 2006; Chen, et al. 2017; Sackman, et al. 2019) whereby the effective population size, N_e , is used to accommodate various deviations from the ideal WF population. Note that N , and hence N_e , are imposed externally on the model as in the later Moran model (2008).

At about the same time, JBS Haldane introduced the branching process to model genetic drift (Haldane 1927; Haldane 1932). In the Haldane model, each gene copy is independently transmitted to K descendants with the mean and variance of $E(K)$ and $V(K)$. Again, we present the haploid model, noting some added complexities with diploidy.

The branching process of the Haldane model can generate N internally by formulating N of the next generation as $E(N') = N \times E(K)$. Since Haldane's model (1927) resets N' to a constant N , a Generalized Haldane (GH) model is proposed here that permits varying N under ecological regulation. Indeed, $E(K)$ is often negatively correlated with N in actual populations (Smith and Slatkin 1973; Sibly and Hone 2002). The GH model thus tracks gene frequency changes as well as regulates N . In the Haldane model, $E(K)$ would determine the trajectory of N changes and $V(K)$ would determine the strength of genetic drift. Thus, there is no drift if $V(K) = 0$ irrespective of N .

The GH model is then applied to several paradoxes of genetic drift (see the overview below). It is possible that mathematical modifications of the conventional models (i.e., WF and Moran models) may resolve these paradoxes as well. Nevertheless, the key question is whether these modifications, often highly sophisticated, are biologically feasible (see Discussion). In comparison,

the Haldane model is intuitively appealing and deserves to be considered as an alternative approach to genetic drift. As we study the ever more complex biological systems in the genomic era, such an alternative would seem desirable.

Results

Overview of the paradoxes of genetic drift

Among the paradoxes, a most curious one is genetic drift in relation to changing population size. The WF model dictates stronger drift when N is smaller. However, when N is very small (say, $N = 10$ in a bacteria or yeast population) and increases to 20, 40, 80 and so on, there is in fact little drift in this exponential phase. Drift will intensify when N grows to near the carrying capacity. This trend is the exact opposite of the standard view. The second paradox concerns the genetic drift of sex-linked genes in relation to sex-dependent breeding successes (Charlesworth and Charlesworth 2000 [\[1\]](#); Bachtrog 2013 [\[2\]](#); Cortez, et al. 2014 [\[3\]](#); Wilson Sayres, et al. 2014 [\[4\]](#); Makova, et al. 2024 [\[5\]](#)). A third paradox is about drift strength when selection is also at work. A new mutation with a selective advantage would always be fixed if there is no random drift. With drift, fixation is probabilistic, but the probability does not depend on N or N_e . This curious property echoes the view that N is a scaling factor of drift that is determined by $V(K)$ (see Results).

The power of the Haldane model in resolving these paradoxes is the focus of this study. Furthermore, the companion study (Wang et al. 2024 [\[6\]](#)) addresses a fourth paradox – the evolution of multi-copy gene systems, whereby evolution proceeds in two stages - between as well as within individuals. Multi-copy gene systems including viruses, mitochondria, transposons and ribosomal genes do not have easily definable N_e . Broadly speaking, even diploidy is also multi-copy systems (Silver 1985 [\[7\]](#); Wu, et al. 1988 [\[8\]](#); Wu, et al. 1989 [\[9\]](#); Lindholm, et al. 2016 [\[10\]](#); Courret, et al. 2019 [\[11\]](#)).

On the Haldane model of genetic drift

In the original Haldane model, genetic drift is defined by the branching process (see Eq. (4) [\[12\]](#) of Chen, et al. (2017) [\[13\]](#)). Let K be the number of progenies receiving the gene copy of interest from a parent; K would follow the same distribution for all neutral variants. In haploids, K is equivalent to the progeny number of each parent. In diploids, K is the transmission success of each gene copy, which is separately tracked. By the Taylor approximations of the ratio of two variables (Kendall, et al. 2006 [\[14\]](#)), we obtain the approximation as

$$V(x) \sim \frac{x(1-x)}{N} \frac{V(K)}{E^2(K)} \quad (2)$$

where x is the variant frequency (see Supplementary Information). This is the same approximation, when $V(K)$ is not very large, as obtained by the WF model (Kimura and Crow 1964; see Chen et al. 2017 [\[13\]](#) for details). For Eq. (2) [\[15\]](#), we provide further extensive simulations on the accuracy in both the fixation probability and fixation time of neutral variants (see Supplementary Information). Note that N , constant or variable, is supplied externally to the WF model but it is tracked internally in the Haldane model (see the next section). The apparent equivalence of Eq. (2) [\[16\]](#) makes a simple point that, between the WF and Haldane models, the mathematical results are often shared but the biological meanings may not be identical. (Similarly, the diffusion and coalescence theories are also distinct approaches to the same population genetic phenomena.)

In the Haldane model, there would be no genetic drift if $V(K) = 0$, regardless of the value of N . When $E(K) = 1$ and N stays constant, we obtain

$$N_e = \frac{NE^2(K)}{V(K)} = \frac{N}{V(K)} \quad (3)$$

Eq. (2) and Eq. (3) have been obtained for the WF model (Kimura and Crow 1963; Crow and Denniston 1988; Chen, et al. 2017). The issue will be highly significant when sex-dependent drift is analyzed.

In Table 1, the data of progeny number from the literature indeed show over-dispersion in all taxa surveyed, i.e., $V(K) > E(K)$. In fact, $V(K) > 10 E(K)$ can often be found, e.g., the ratio of $V(K)/E(K)$ among males of the mandrill (*Mandrillus sphinx*), the great reed warbler (*Acrocephalus arundinaceus*), and the rhesus macaque (*Macaca mulatta*) at 19.5, 12.6 and 11.3, respectively (Hasselquist 1995; Setchell, et al. 2005; Dubuc, et al. 2014). We have also found that $V(K)$ may be far larger than $E(K)$ in other biological systems such as viruses (Ruan, Luo, et al. 2021; Ruan, Wen, et al. 2021; Hou, et al. 2023).

I. The first paradox – The paradox of changing N

1. Empirical demonstration and simulation

This “paradox of changing N ” is that “genetic drift increases when N increases” in direct contradiction with Eq. (2). Fig. 1a shows the results in a cell culture in the exponential growth phase where each cell doubles every 13 hours. Hence, $V(K) \sim 0$ with almost no drift when N is as small as < 50 . Genetic drift is expected to increase as N approaches the carrying capacity.

The paradox is exemplified through computer simulations in Fig. 1b-c. These panels show the drift pattern in the Haldane model and the standard WF model, respectively, when the populations are growing as the logistic growth model (see Methods). In the WF model, the drift is strong initially but weakened as N increases. The pattern is reversed in the Haldane model. The contrast between the two panels is the paradox.

To verify the simulations of Fig. 1b-c, we study the growth of yeast populations under laboratory conditions where cells appear to increase following the logistic growth curve. As shown in Fig. 1d, the lower portion of the S-curve portrays the exponential growth in a low-density population, while the upper curve indicates a slowdown in growth near the population’s carrying capacity. We directly recorded the individual output (K) of each yeast cell in the early ($n = 25$) and late ($n = 65$) stage of population growth under real-time high-content fluorescence imaging (Methods).

It’s evident that $V(K)$ is decoupled from $E(K)$. In the early stage, $E(K)$ is 10.04 with $V(K) = 0.60$ over five one-hour intervals. In the high-density phase, $E(K)$ decreases to 2.83 but $V(K)$ increases to 1.09 in four intervals (Fig. 1e-f). Most interestingly, during the late stage of population growth, $V(\Delta K)$ (adjusted to the four-hour time span for comparison; see Methods) indeed increases as N increases (Supplementary Table 3). At the last time point, the population is closest to the carrying capacity and $V(\Delta K)$ experiences a substantial leap from around 1.12 to 1.82. Overall, the variance in progeny number $V(K)$ in a high-density population is always greater than in a low-density population (Fig. 1g). Indeed, $V(K)$ may exhibit an increase with the increase in N , sometimes outpaces the latter.

Measurements of these two growth phases demonstrate the “paradox of changing N ” by showing i) there is almost no drift when N is very small; ii) as N increases, $V(K)$, hence, genetic drift would also increase. Furthermore, $V(K)$ may increase sharply when N is close to the carrying capacity. This trend appears to echo the field observations by Coltman, et al. (1999) who show that $V(K)$ increases more than 5-fold when N increases only 2.5-fold in the United Kingdom (UK) population of Soay rams (*Ovis aries*).

Table 1.

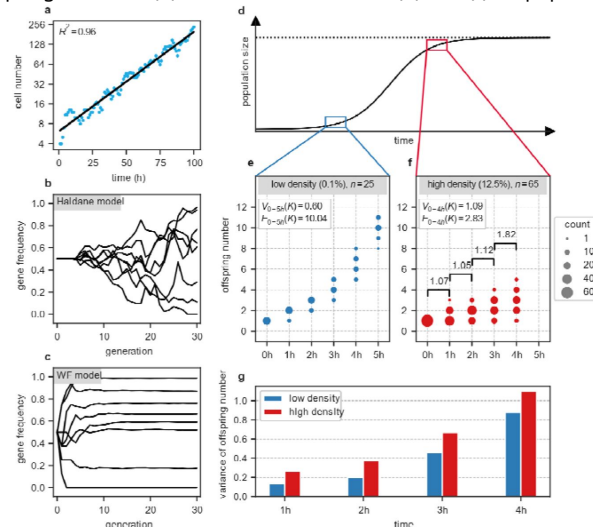
Field record of $E(K)$ and $V(K)$ across diverse taxa.

Species	Type	Order	Gender	$E(K)$	$V(K)$	$V(K)/E(K)$	Sample Size	Literature
<i>Macaca mulatta</i>	Mammal	Primates	Female	7.7	15.4	2.0	132	Dubuc, et al. (2014)
			Male	8.7	98.7	11.3	86	
<i>Mandrillus sphinx</i>	Mammal	Primates	Female	4.7	19.8	4.2	51	Setchell, et al. (2005)
			Male	3.5	68.2	19.5	53	
<i>Macaca sylvanus</i>	Mammal	Primates	Female	1.2	2.9	2.4	98	Kuester, et al. (1995)
			Male	1.5	3.9	2.7	81	
<i>Peromyscus californicus</i>	Mammal	Rodentia	Female	4.7	7.7	1.6	16	Ribble (1992)
			Male	4.4	13.7	3.1	20	
<i>Turdus merula</i>	Bird	Passeriformes	Female	4.9	23.4	4.8	80	Wysocki, et al. (2019)
			Male	4.0	27.0	6.8	72	
<i>Acrocephalus arundinaceus</i>	Bird	Passeriformes	Female	7.7	53.3	6.9	82	Hasselquist (1995)
			Male	10.5	132.3	12.6	64	
<i>Charadrius nivosus</i>	Bird	Charadriiformes	Female	1.7	5.3	3.1	105	Herman and Colwell (2015)
			Male	2.2	9.0	4.1	90	
<i>Actitis macularia</i>	Bird	Charadriiformes	Female	5.2	26.9	5.1	66	Oring, et al. (1991)
			Male	3.3	17.2	5.2	103	
<i>Buteo buteo</i>	Bird	Accipitriformes	Female	3.5	13.8	4.0	106	Krüger and Lindström (2001)
			Male	2.7	9.3	3.4	132	

Fig. 1.

The paradox of genetic drift when the population size (N) changes.

(a) In the laboratory cell culture, nearly all cells proliferate when N is very small (interpreted in the next panel). (b) The simulation of Haldane model shows little genetic drift in the exponential phase. As in (a), drift may increase due to the heightened competition as the population grows. (c) Simulation by the WF model shows a pattern of drift opposite of the Haldane model. (d) The patterns of drift at the low and high N are analyzed in the framework of the logistic growth model. (e–f) Measurements of genetic drift in laboratory yeast populations at low and high density as defined in (d). The progeny number of each cell, K , is counted over 4 or 5 intervals as shown by the dots, the sizes of which reflect the cell number. $E(K)$ and $V(K)$ are presented as well. In panel (f), the change of offspring number overtime, denoted as $V(\Delta K)$, is shown above the braces. (g) The variance of offspring number $V(K)$ increases, observed in (e) and (f), as population size increases.



2. The density-dependent Haldane (DDH) model – A first attempt at the Generalized Haldane model

To explain the paradox of changing N , the model has to track and regulate N . Ironically, although the branching process can track the changes, the process has been deemed unsuitable for population genetics. This is because $N_t = N_0 E(K)^t$ would drive N to approach either zero or infinity as t becomes very large. Indeed, Haldane's original model reset N' to N in each generation. Any generalized Haldane (GH) model must take advantage of this feature by regulating N via $E(K)$, as attempted in this section.

The “paradox of changing N ” can be explained by $N_e = N/V(K)$ if the numerator and denominator move in the same direction. Most interestingly, $V(K)$ may increase more than N itself. This has been shown when $V(K)$ is close to zero while N is growing exponentially. In addition, when N is very near the carrying capacity, even a small increase in N would intensify the competition and inflate $V(K)$. This is shown in the yeast experiment of [Fig. 1d](#) and supported by [Coltman, et al. \(1999\)](#) field study of ram populations in UK. Thus, the relationship between N and $V(K)$ may realistically lead to this paradox.

We now extend the Haldane model by incorporating density-dependent regulation of N in order to define the conditions of this paradox. The model developed here is on an ecological, rather than an evolutionary time scale. The simplest model of N regulation is the logistic equation:

$$\frac{dN}{dt} = rN \frac{C_k - N}{C_k} \quad (4)$$

with C_k being the carrying capacity. In the ecological time scale, a changing N would mean that the population is departing from C_k or moving toward it ([Fig. 2a](#)). (As will be addressed in Discussion, changing N in the WF model is depicted in [Fig. 2b](#) whereby N is at C_k . C_k may evolve too, albeit much more slowly in an evolutionary time scale.) Here, we consider changes in N in the ecological time scale. Examples include the exponential growth when $N \ll C_k$, seasonal fluctuation in N ([Frankham 1995](#); [Charlesworth 2009](#)) and competition modeled by the Lotka-Volterra equations ([Bomze 1983](#)).

Details of the DDH model are given in the Supplementary Information. A synopsis is given here: We consider a non-overlapping haploid population with two neutral alleles. The population size at time t is N_t . We assume that expected growth rate $E(K)$ is greater than 1 when $N < C_k$ and less than 1 when $N > C_k$, as defined by [Eq. \(5\)](#) below:

$$E(K_t) = \frac{E(N_{t+1})}{N_t} = \frac{r + 1}{1 + r \left(\frac{N_t}{C_k}\right)^z} \quad (5)$$

The slope of $E(K)$ vs. N (i.e., the sensitive of growth rate to changes in population size), as shown in [Fig 3a](#), depends on z . To determine the variance $V(K)$, we assume that K follows the negative binomial distribution whereby parents would suffer reproduction-arresting injury with a probability of p_t at each birthing (Supplementary Information). Accordingly, $V(K)$ can then be expressed as

$$\frac{V(K_t)}{E(K_t)} = \frac{1 - p_t}{p_t} = a(N_t)^b \quad (6)$$

By [Eq. \(6\)](#), the ratio of $V(K)/E(K)$ could be constant, decrease or increase with the increase of population size. With $E(K)$ and $V(K)$ defined, we could obtain the effective population size by substituting [Eq. \(5\)](#) and [Eq. \(6\)](#) into [Eq. \(3\)](#).

$$N_e = \frac{N_t E^2(K_t)}{V(K_t)} = \frac{N_t (r + 1)}{a(N_t)^b \left[1 + r \left(\frac{N_t}{C_k}\right)^z\right]} \quad (7)$$

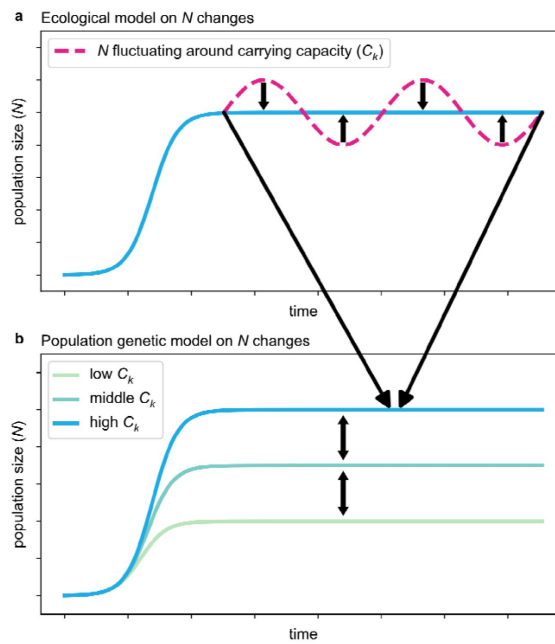


Fig. 2.

The meaning of population size (N) changes in ecology vs. in population genetics.

(a) In ecology, changing N would generally mean a population approaching or departing the carrying capacity. (b) In population genetics, a population of size N is assumed to be at the carrying capacity, C_k . Thus, changes in N would mean an evolving C_k , likely the consequence of environmental changes. The arrows indicate the disparity in time scale between the two scenarios.

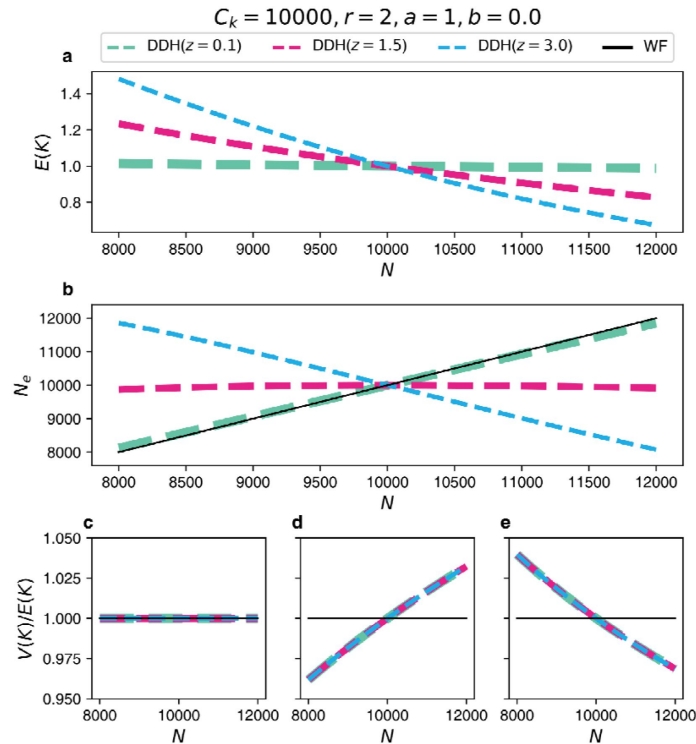


Fig. 3.

Genetic drift as a function of population size in the DDH model.

For all panels, the carrying capacity is $C_k = 10,000$ and the intrinsic growth rate is $r = 2$. **(a)** When N increases, $E(K)$ decreases as modeled in [Eq. \(5\)](#). The z value of [Eq. \(5\)](#) (0.1, 1.5 and 3) determines the strength of N regulation, indicated by the slope of $E(K)$ near $C_k = 10,000$. **(b)** Depending on the strength of N regulation near C_k , genetic drift can indeed decrease, increase or stay nearly constant as the population size increases. Thus, the conventional view of N_e being positively dependent on N is true only when the regulation of N is weak (the green line). At an intermediate strength (the red line), N_e is nearly independent of N . When the regulation becomes even stronger at $z = 3$, N_e becomes negatively dependent on N . **(c-e)** $V(K)/E(K)$ of [Eq. \(6\)](#) is shown as a function of N . The results of panel (b) are based on a constant $V(K)/E(K)$ shown in panel (c). Interestingly, the results of panel (b) would not be perceptibly changed when $V(K)/E(K)$ varies, as shown in panel (d) and (e).

Eq. (7) presents the relationship between effective population size (N_e) and the population size (N) as shown in Fig. 3. The density-dependent $E(K)$ could regulate N with different strength (Fig. 3a). The steeper the slope in Fig. 3a, the stronger the regulation.

The main results of this study are depicted in Fig. 3b. First, with no or little regulation of N , N_e and N are strongly correlated. The green dashed lines portray the conventional view of decreasing drift with increasing N . Second, under a particular strength of N regulation, the red line shows no dependence of N_e on N , meaning that genetic drift is independent of N . Finally, as N becomes strongly regulated, N_e and N would be negatively correlated as the blue dashed line shows. This trend is the paradox of changing N .

In summary, genetic drift effect can indeed decrease, increase or stay nearly constant as the population size increases, depending on the strength of N regulation near C_k . We further note that the $V(K)/E(K)$ ratio of Eq. (7) can be independent (Fig. 3c), positively dependent (Fig. 3d), or negatively dependent (Fig. 3e) on N . Interestingly, the results of Fig. 3b are nearly identical when the $V(K)/E(K)$ ratio changes (Supplementary Figs 1-3). These results show that the strength of genetic drift depends on the ecology that governs $E(K)$, $V(K)$ and N .

The paradox of changing N may not be an exception but a common rule in natural populations. Note that the WF approximation yielding Eq. (2) assumes nearly constant N . The assumption would imply very strong N regulation near C_k , which is precisely the condition leading to the paradox of changing N (Fig. 3). Coltman et al. (1999)'s observations of the reproduction in rams is such a case.

II. The paradox of genetic drift in sex chromosomes

Since the relative numbers of Y, X and each autosome (A) in the human population are 1:3:4, the WF model would predict the genetic diversity (θ_Y , θ_X and θ_A , respectively) to be proportional to 1:3:4. In a survey of human and primate genetic diversity, θ_Y is almost always less than expected as has been commonly reported (Hammer, et al. 2001; Wang, et al. 2014; Wilson Sayres, et al. 2014; Makova, et al. 2024). The low θ_Y value has been used to suggest that human Y chromosomes are under either positive or purifying selection (Wang, et al. 2014; Wilson Sayres, et al. 2014; Makova, et al. 2024). Similarly, the reduced diversity X-linked genes was interpreted as a signature of positive selection (Pan, Liu, et al. 2022).

As pointed out above, under-estimation of genetic drift is a major cause of over-estimation of selective strength. Our goal is hence to see if genetic drift of different strength between sexes can account for the observed genetic diversities. Details will be presented in Wang et al. (2024). Below is a synopsis.

Let V_m and V_f be the $V(K)$ for males and females respectively and $\alpha' = V_m/V_f$ is our focus. (We use α' for the male-to-female ratio in $V(K)$ since α is commonly used for the ratio of mutation rate (Miyata, et al. 1987; Makova and Li 2002).) The three ratios, θ_Y/θ_X (denoted as R_{YX}), θ_Y/θ_A (R_{YA}) and θ_X/θ_A (R_{XA}), could be expressed as the functions of α' , which incorporate the relative mutation rates of autosomes, X and Y (Supplementary Information). These relative mutation rates can be obtained by interspecific comparisons as done by Makova and Li (2002).

We note that there are many measures for the within-species genetic diversity, $\theta = 4N_e\mu$. Under strict neutrality and demographic equilibrium, these measures should all converge. In the neutral equilibrium, the infinite site model dictates the frequency spectrum to be $\xi_i = \theta/i$, where ξ_i is the number of sites with the variant occurring i times in n samples. Since every frequency bin is a measure of θ , different measures put different weights on the i -th bin (Fay and Wu 2000; Fu 2022). While π , the mean pairwise differences between sequences, is most commonly used in

the literature, we use several statistics to minimize the possible influences of selection and demography (Wang et al. 2024). In this synopsis, we used the Watterson estimator (Watterson 1975) as the measure for θ_Y , θ_X and θ_A by counting the number of segregating sites.

Using any of the three ratios (θ_Y/θ_X , θ_Y/θ_A and θ_X/θ_A), we can obtain α' ; for example, $R_{YA} = \theta_Y/\theta_A$ and, hence,

$$R_{YA} = \frac{y}{4} \frac{1 + \alpha'}{2\alpha'}$$

where y is the mutation rate of Y-linked sequences relative to autosomal sequences. With rearrangement,

$$\alpha' = \frac{y}{8R_{YA} - y} \quad (8)$$

Similar formulae of α' can be obtained for R_{YX} and R_{XA} but the accuracy for estimating α' is highest by R_{YA} whereas R_{XA} is the least accurate for this purpose (Supplementary Information).

Table 2 presents the α' estimates in chimpanzees and bonobos. This is part of a general survey in mammals with a strong emphasis on primates and humans (Wang et al. 2024). It is almost always true that $\alpha' > 5$ in primates. Sources of data used are also given in **Table 2**. As shown for chimpanzees, α' is often far larger than 5, above which the resolution is very low as can be seen in **Eq. (8)**. Note that, when R_{YA} is under-estimated and approaching $y/8$, α' would increase rapidly when the denominator is close to 0. That is why α' often becomes infinity (**Supplementary Fig. 5**). The estimated α' (MSE) in **Table 2** alleviates the problem somewhat by using all three ratios to calculate the mean square error (MSE) (Supplementary Information).

Among primates surveyed, bonobo is the only exception with $\alpha' < 1$. While chimpanzees and bonobos are each's closest relatives, their sexual behaviors are very divergent (de Waal 1995; De Waal and Lanting 2023). With unusually strong matriarchs, bonobo society seems to stand out among primates in sexual dominance. The α' value is important in behavioral studies and, particularly, in primatology and hominoid research. If 0.1% of males sire 100 children and the rest have the same K distribution as females, V_m/V_f would be ~ 10 . Such outlier contribution is the equivalent of super-spreaders in viral evolution which may easily be missed in field studies. In this brief exposition, we highlight again that $V(K)$ or, more specifically, V_m and V_f are key to genetic drift. The differences in drift strength among X, Y and autosomes constitute a curious (but not immediately recognizable) paradox for the WF models as will be addressed in Discussion.

III. The paradox of genetic drift under selection

Genetic drift operates on neutral as well as non-neutral mutations. Let us assume a new mutation, M, with a frequency of $1/N$ is fixed in the population with the probability of P_f . Fisher (1930) first suggested that the fixation probability of an advantageous mutation should increase in growing populations, while decrease in shrinking populations. If there is no genetic drift, a beneficial mutation will always be fixed and $P_f = 1$. In the WF model, it is well known that $P_f \sim 2s$ for a new advantageous mutation, with fitness gain of s , while population size is large (Haldane 1932; Kimura 1962). This seems paradoxical that the determinant of genetic drift, $1/N_e$, does not influence P_f (Otto and Whitlock 1997; Lanfear, et al. 2014).

In the Method section, we show that P_f under the Haldane model is given by

$$P_f = \sum_{t=0}^{\infty} \left[1 - (u_M(t))^{M_0} \right] \left[(u_W(t))^{W_0} - (u_W(t-1))^{W_0} \right] \quad (9)$$

	$\theta_Y (\times 10^{-3})$	$\theta_X (\times 10^{-3})$	$\theta_A (\times 10^{-3})$	$\alpha' \text{ (MSE)}$
Chimpanzee				
Western	0.034	0.533	1.623	4.98
Eastern	0.144	1.044	1.446	25.90
Nigeria-Cameroon	0.213	0.818	0.703	Inf ^a
Bonobo	0.492	0.446	0.867	0.57

Data for counting the segregating sites are from Hallast, et al. (2016) and Nam, et al. (2015).

θ_Y , θ_X , θ_A : genetic diversity for Y chromosome, X chromosome, and Autosomes (A), measured by Watterson estimator (Watterson 1975).

α' (MSE): optimal estimate for α' based on the minimum mean square error (MSE).

^a Inf (infinity) designates cases when the estimated α' based on MSE is greater than 1000 (i.e., $V_m \gg V_f$).

Table 2.

Estimation of V_m/V_f in chimpanzee and bonobo.

where M_0 and W_0 are the initial number of mutant allele (M allele) and wildtype allele (W allele), with $N = M_0 + W_0$. We note that $u_M(t)$ and $u_W(t)$, respectively, are the extinction probabilities of M allele and W allele by generation t with the initial number of alleles of 1. While obtaining the direct analytical solution of Eq. (9) may not be feasible, we could obtain its numerical solution due to its convergence as t increases (see Methods). The accuracy of the numerical solution from Eq. (9) is confirmed through simulation (Supplementary Fig. 4). Moreover, with the aid of the WF model (see Supplementary Information), we could obtain the approximation of Eq. (9) as follows.

$$P_f = \frac{2s}{V(K)} \quad (10)$$

We verify Eq. (10) by both the numerical solution from Eq. (9) and simulations based on the branching process (Supplementary Fig. 4). The fixation probabilities obtained by numerical solution vs. those inferred from Eq. (10) are shown in Fig. 4. The salient feature is that the fixation probability of $2s$, as in the classical formula, would be a substantial over-estimate when $V(K)$ is larger than $E(K)$. Eq. (10) is sufficiently accurate as long as $N \geq 50$. When N is as small as 10, the theoretical result is biased. Indeed, at such a low N value, the population is prone to extinction. The DDH model presented above should rectify this deficiency.

The main message of Eq. (10) is that genetic drift under positive selection is influenced by s and $V(K)$ but not by N . This independence from N is explicable: when an advantageous mutation increases to a certain level still far below N (depending mainly on s), its fixation would be almost certain. This may be a most direct argument against equating genetic drift with N , or N_e (which is a function of changing N 's).

Discussion

Genetic drift should broadly include all random forces affecting evolutionary trajectories. Hence, when N is not regulated by the models themselves but supplied externally, some random forces are excluded from the modeling. The DDH model may be a first-generation GH model that incorporates N regulation. We note that the original Haldane model suppresses N fluctuation and is hence a special case of DDH with extremely strong N regulation near C_k .

We shall first clarify the first paradox, the “paradox of changing N 's”. This paradox is in the ecological time scale whereby N is either growing or oscillating around the carrying capacity (Fig. 2a). In this time scale, drift may often (but certainly not always) increase in strength as N increases. While N and N_e are often poorly correlated in WF models (Crow and Kimura 1970; Charlesworth 2009; Lynch, et al. 2016), there is no demonstration how N and N_e can be negatively correlated in the absence of N regulation. Nevertheless, the WF models do work well in the evolutionary time scale (Fig. 2b). For example, by the PSMC model (Li and Durbin 2011), both the European and Chinese populations experience a severe bottleneck 10-60 kyr ago. Presumably, various environmental forces may have reduced C_k drastically. Averaged over the long-time span, N should be at or near C_k . In short, the N value in the WF models is both N and C_k at the evolutionary time scale. It should also be mentioned that the DDH model is distinct from the concept of “genetic draft” that involves selection and hitchhiking (Gillespie 2000, 2001).

The second paradox of sex-dependent drift is about different $V(K)$'s between sexes (generally $V_m > V_f$) but the same $E(K)$ between them. In the conventional models of sampling, it is not clear what sort of biological sampling scheme could yield $V(K) \neq E(K)$, let alone two separate $V(K)$'s with one single $E(K)$. Mathematically, given separate K distributions for males and females, it is unlikely that

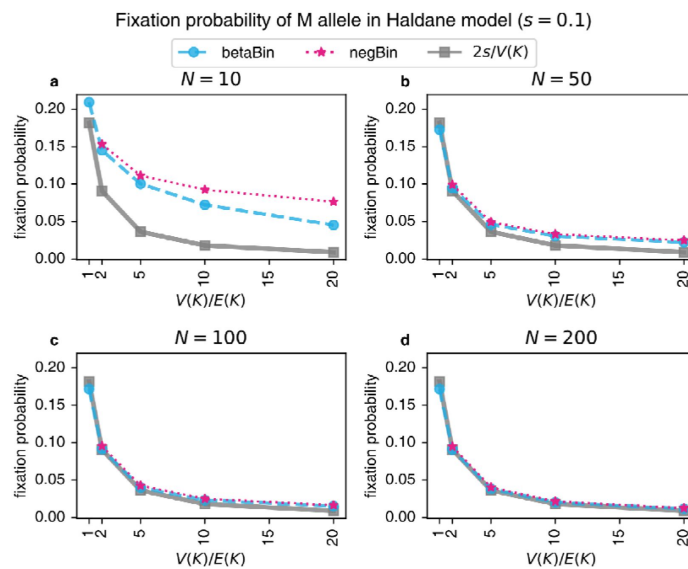


Fig. 4.

Fixation probability of a new advantageous mutation in the Haldane model.

The fixation probabilities of a new advantageous mutation with the selective advantage of $s = 0.1$ are calculated based on approximate solution from [Eq. \(9\)](#) (i.e., $2s/V(K)$) as well as numerical solution from [Eq. \(10\)](#). The numerical solution from [Eq. \(10\)](#) has been confirmed accurate by simulations (Supplementary Fig. 4). **(a-b)** When $N < 50$, the approximate fixation probability (the gray line) is lower than the simulated values (the color lines) due to population extinction. **(c-d)** By the Haldane model, the expected fixation probability of [Eq. \(9\)](#) is accurate when N reaches 100, as in most natural populations.

$E(K)$ for the whole population could be 1, hence, the population would either explode in size or decline to zero. In short, N regulation has to be built into the genetic drift model as the GH model does to avoid this paradox.

The third paradox of genetic drift is manifested in the fixation probability of an advantageous mutation, $2s/V(K)$. As explained above, the fixation probability is determined by the probability of reaching a low threshold that is independent of N itself. Hence, the key parameter of drift in the WF model, N (or N_e), is missing. This paradox supports the assertion that genetic drift is fundamentally about $V(K)$ with N being a scaling factor. Note the absence of genetic drift at all N 's when $V(K) = 0$.

The fourth paradox is about the multi-copy gene systems such as viruses and rRNA genes covered in the companion study (Wang, et al. 2024 [\[1\]](#)). These systems evolve both within and between hosts. Given the small number of virions transmitted between hosts, drift is strong in both stages as shown by the Haldane model (Ruan, Luo, et al. 2021 [\[2\]](#); Ruan, Wen, et al. 2021 [\[3\]](#); Hou, et al. 2023 [\[4\]](#)). Therefore, it does not seem possible to have a single effective population size to account for the genetic drift in two stages with very different biological processes. The inability to deal with multi-copy gene systems may explain the difficulties in accounting for the SARS-CoV-2 evolution (Deng, et al. 2022 [\[5\]](#); Pan, Liu, et al. 2022 [\[6\]](#); Ruan, et al. 2022 [\[7\]](#); Hou, et al. 2023 [\[4\]](#); Ruan, et al. 2023 [\[8\]](#)).

As the domain of evolutionary biology expands, many new systems do not have definable populations that fit the criteria of WF populations (such as panmixia in mating or dispersal). Multi-copy gene systems are obvious examples. Others include domestications of animals and plants that are processes of rapid evolution (Diamond 2002 [\[9\]](#); Larson and Fuller 2014 [\[10\]](#); Purugganan 2019 [\[11\]](#); Chen, Yang, et al. 2022 [\[12\]](#); Pan, Zhang, et al. 2022 [\[13\]](#); Wang, et al. 2022 [\[14\]](#)). Due to the very large $V(K)$ in domestication, drift must have played a large role. Somatic cell evolution is another example with “undefinable” genetic drift (Wu, et al. 2016 [\[15\]](#); Chen, et al. 2017 [\[16\]](#); Chen, et al. 2019 [\[17\]](#); Ruan, et al. 2020 [\[18\]](#); Chen, Wu, et al. 2022 [\[19\]](#)). The Haldane model is an individual-output model (Chen, et al. 2017 [\[16\]](#)) whereby the collection of individuals constitute the “population”.

We understand that further modifications of the WF models may account for some or all of these paradoxes. However, such modifications have to be biologically feasible and, if possible, intuitively straightforward. Such possible elaborations of WF models are beyond the scope of this study. We are only suggesting that the Haldane model can be extensively generalized to be an alternative approach to genetic drift. The GH model attempts to integrate population genetics and ecology and, thus, can be applied to genetic systems far more complex than those studied before. The companion study is one such example.

Methods

Cell culture and image analysis

NIH3T3 cells, a fibroblast cell line that was isolated from a mouse NIH/Swiss embryo, were stably transfected with the fluorescent, ubiquitination-based cell cycle indicator (Fucci) (Sakaue-Sawano, et al. 2008 [\[20\]](#)) plasmid using Lipofectamine 3000 Transfection Reagent (Invitrogen) following the manufacturer's specified instructions. The Fucci-labeled cells exhibited distinct fluorescent signals indicative of G1, S, and G2/M phases, represented by red, yellow, and green, respectively. NIH3T3-Fucci cell was derived from single-cell colony and cultured in DMEM supplemented with 10% Calf Bovine Serum and penicillin/streptomycin. All cells were maintained at 37°C with 5% CO₂. Subsequently, the cells underwent extended time-lapse imaging using high-content fluorescence

microscopy (PerkinElmer Operetta CLS) equipped with a 10x objective lens. Images were captured hourly over a 100-hour period, and the analysis was conducted using ImageJ (Fiji) (Schneider, et al. 2012) to count the number of cells (Supplementary Table 1).

Yeast strain construction

Strains were constructed on the genetic background of *Saccharomyces cerevisiae* strain BY4741. A GFP or BFP fluorescent protein expression cassette, under the control of the TDH3 promoter, was inserted into the pseudogene locus YLL017W. Transformations followed a published protocol (Gietz and Schiestl 2007). Transformants were plated on synthetic complete medium without uracil (SC-Ura), and from these, single colonies were selected. Confirmation of replacements was achieved through PCR, and cassette verification was performed using fluorescence microscopy. Subsequently, the constructed strains were cultivated non-selectively in YPD medium (1% Yeast extract, 2% Peptone, 2% D-glucose) at 30°C on a rotary shaker.

Estimation of $V(K)$ and $E(K)$ in yeast cells

To discern division events, even under high concentration, we conducted co-cultures of the GFP-yeast and BFP-yeast at ratios of 1:1 and 1:25, with the initial cell concentration of 0.1% and 12.5%, respectively. Then the yeast cells were then continuously imaged under high-content fluorescence microscopy (PerkinElmer Operetta CLS) for 10 hours with 1-hour intervals to observe the individual offspring of GFP-yeast. Yeast cells with a distinct offspring number (K) within the initial 5 hours (with the high-density group limited to the first 4 hours) were documented (Supplementary Table 2). Subsequently, the mean and variance of K for each cell were calculated across various time intervals as follows.

According to the law of total variance,

$$V(I_t) = E[V(I_t|I_{t-1})] + V[E(I_t|I_{t-1})] = V_{t-1,t}E[I_{t-1}] + E_{t-1,t}^2V[I_{t-1}]$$

where I_t is the offspring number at time t for an initial single cell (i.e., the total number of progeny cells for a single cell after t hours), as documented in Supplementary Table 2. And $E_{t-1,t}$ represents the average of offspring number after single time interval (1 hour here) for each single cell from time $t-1$ to time t , while $V_{t-1,t}$ is the corresponding variance. Utilizing the documented total cell count at time t (denoted by N_t), we could calculate the average of offspring number for a single yeast cell in a specific time interval, e.g., $E_{t-1,t} = N_t / N_{t-1}$ and $E[I_{t-1}] = N_t / N_0$. With some rearrangement,

$$V_{t-1,t} = \frac{V(I_t) - E_{t-1,t}^2V[I_{t-1}]}{E[I_{t-1}]} = \frac{V(I_t) - (N_t/N_{t-1})^2V[I_{t-1}]}{N_{t-1}/N_0}$$

Applying the aforementioned equations, we observed that both $V_{t-1,t}$ (i.e., $V(K)$ within a one-hour interval) and $V(I_t)$ (i.e., the $V(K)$ from 0 to time t) for yeast cells under high density (12.5%) are generally greater than the values under low density (Supplementary Table 3). This suggests that the variance of offspring number in high-density conditions could be larger than that in a low-density context. It's noteworthy that the estimated value of $V_{t-1,t}$ could be less than zero (Supplementary Table 3), implying a very small variance in the number of offspring. Furthermore, the occurrence of negative values for $V_{t-1,t}$ is more frequent under low density than high density, indicating the higher variance of offspring number in high density as previously suggested.

To show the variance of offspring number overtime, we also calculated the change of offspring number within successive one-hour intervals, denoted as $V(I_t - I_{t-1})$. This value is intricately linked to the variance of offspring overtime. To facilitate the comparison with the total variance from 0h to 4h, we set $V(\Delta K) = 4V(I_t - I_{t-1})$ to account for the four-hour time span.

Simulation of genetic drift in the Haldane model and the Wright-Fisher (WF) model

In both models, interactions between individuals are implicitly included through the dependency of the average number of offspring on population size, as defined by Eq. (5) [\[1\]](#). This dependency leads to the logistic population growth, reflecting the density-dependent interactions. The initial population size is set to 4, with initial gene frequency (i.e., the frequency of mutant allele) of 0.5. And the carrying capacity is established at 100,000, with $r = 1$ and $z = 1$. In WF model, the population size at next generation is $N_{t+1} = N_t \times E(K_t)$. Note if the calculated value of N_{t+1} is not an integer, we round it to the nearest whole number. The number of mutant alleles follows a binomial distribution, allowing the simulation of gene frequency overtime in WF model. For Haldane model, the number of offspring number is assumed to follow a beta-binomial distribution. The mean of offspring number $E(K_t)$ is obtained from Eq. (5) [\[1\]](#). The variance $V(K_t)$ is obtained from Eq. (6) [\[1\]](#) with parameters $a = 1/300$ and $b = 1.2$. Given the mean and variance of the beta-binomial distribution, we simulated the number of mutant alleles and population size overtime and then traced the gene frequency overtime.

Fixation probability of a new mutation in Haldane model

Here we obtain the fixation probability of advantageous mutation in Haldane model governed by a branching process. The Haldane model considers a well-mixed population of N_t haploid parents with only two types of alleles (W for wildtype, M for mutant). There is no migration and new mutations in this model. Each individual is assumed to independently reproduce in discrete and non-overlapping generation, $t = 0, 1, 2, \dots$. Thus, the number of offspring per allele (also an individual in this haploid population) at any generation is represented by a set of identically and individually distributed random variables. In particular, the numbers of offspring of M and W alleles (denoted as K_M and K_W respectively) can be represented by following distributions.

$$\begin{aligned} P(K_M = k) &= i_k \text{ for } k = 0, 1, 2, \dots \\ P(K_W = k) &= j_k \text{ for } k = 0, 1, 2, \dots \end{aligned}$$

For a particular distribution, we can obtain the average number of offspring as follows.

$$\begin{aligned} E(K_M) &= \sum_{k=0}^{\infty} k \times i_k \\ E(K_W) &= \sum_{k=0}^{\infty} k \times j_k \end{aligned}$$

With the selection coefficient of s (0 for neutral mutation, positive for advantageous mutation, and negative for deleterious mutation) of M allele, average offspring number of M allele and W allele will follow the relationship.

$$E(K_M) = E(K_W)(1 + s)$$

Now, the evolution of the number of M allele (denoted as M_t) and the number of W allele (denoted as W_t) as time process is a branching process.

To compare Haldane model with WF model (the offspring number follows Poisson distribution with variance and mean equal to 1), we will set $E(K_W) = 1$, $E(K_M) = 1 + s$. And then let the offspring number follow a more general and realistic distribution (including negative binomial distribution (negBin), beta-binomial distribution (betaBin)), which can let the ratio of variance to mean range from 1 to a large number. (For simplicity, we let $V(K_M)/E(K_M) = V(K_W)/E(K_W)$.) Based on the branching process, we obtained the fixation probability M alleles (see Supplementary Information for more details on the derivation).

$$P_f = \sum_{t=0}^{\infty} [1 - (u_M(t))^{M_0}] [(u_W(t))^{W_0} - (u_W(t-1))^{W_0}]$$

where

$$u_M(t) = P(M_t = 0 | M_0 = 1) = \sum_{k=0}^{\infty} P(M_t = 0 | M_{t-1} = k) = \sum_{k=0}^{\infty} i_k (u_M(t-1))^k$$

$$u_W(t) = P(W_t = 0 | W_0 = 1) = \sum_{k=0}^{\infty} P(W_t = 0 | W_{t-1} = k) = \sum_{k=0}^{\infty} j_k (u_W(t-1))^k$$

Note $u_M(t=1) = P(K_M = 0) = i_0$, $u_W(t=1) = P(K_W = 0) = j_0$. And both $\{u_M(t), t = 0, 1, 2, \dots\}$ and $\{u_W(t), t = 0, 1, 2, \dots\}$ are a bounded monotonic sequence. Thus, although we cannot obtain the direct analytical solution for P_f we could easily obtain their numerical solution by iteration to the case when both $u_M(t)$ and $u_W(t)$ converge.

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Additional files

Supplementary Information [↗](#)

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Reviewer #1 (Public review):

The revision by Ruan et al clarifies several aspects of the original manuscript that were difficult to understand, and I think it presents some useful and interesting ideas. I understand that the authors are distinguishing their model from the standard Wright-Fisher model in that the population size is not imposed externally, but is instead a consequence of the stochastic reproduction scheme. Here, the authors chose a branching process but in principle any Markov chain can probably be used. Within this framework, the authors are particularly interested in cases where the variance in reproductive success changes through time, as explored by the DDH model, for example. They argue with some experimental results that there is a reason to believe that the variance in reproductive success does change over time.

One of the key aspects of the original manuscript that I want to engage with is the DDH model. As the authors point out, their equations 5 and 6 are assumptions, and not derived from any principles. In essence, the authors are positing that the variance in reproductive success, given by 6, changes as a function of the current population size. There is nothing "inherent" to a negative binomial branching mechanism that results in this: in fact, the variance in offspring number could in principle be the same for all time. As relates to models that exist in the literature, I believe that this is the key difference: unlike Cannings models, the authors allow for a changing variance in reproduction through time.

This is, of course, an interesting thing to consider, and I think that the situation the authors point out, in which drift is lower at small population sizes and larger at large population sizes, is not appreciated in the literature. However, I am not so sure that there is anything that needs to be resolved in Paradox 1. A very strong prediction of that model is that N_e and N could be inversely related, as shown by the blue line in Fig 3b. This suggests that you could see something very strange if you, for example, infer a population size history using a Wright-Fisher framework, because you would infer a population *decline* when there is in fact a population *expansion*. However, as far as I know there are very few "surprising population declines" found in empirical data. An obvious case where we know there is very rapid population growth is human populations; I don't think I've ever seen an inference of recent human demographic history from genetic data that suggests anything other than a massive population expansion. While I appreciate the authors empirical data supporting their claim of Paradox 1 (more on the empirical data later), it's not clear to me that there's a "paradox" in the literature that needs explaining so much as this is a "words of caution about interpreting inferred effective population sizes". To be clear, I think those words of caution are important, and I had never considered that you might be so fundamentally misled as to infer decline when there is growth, but calling it a "paradox" seems to suggest that this is an outstanding problem in the literature, when in fact I think the authors are raising a *new* and important problem. Perhaps an interesting thing for the authors to do to raise the salience of this point would be to perform simulations under this model and then infer effective population sizes using e.g. dadi or psmc and show that you could identify a situation in which the true history is one of growth, but the best fit would be one of decline

The authors also highlight that their approach reflects a case where the population size is determined by the population dynamics themselves, as opposed to being imposed externally as is typical in Cannings models. I agree with the authors that this aspect of population regulation is understudied. Nonetheless, several manuscripts have dealt with the case of population genetic dynamics in populations of stochastically fluctuating size. For example, Kaj and Krone (2003) show that under pretty general conditions you get something very much like a standard coalescent; for example, combining their theorem 1 with their arguments on page 36 and 37, they find that exchangeable populations with stochastic population dynamics where the variance does not change with time still converge to exactly the coalescent you would expect from Cannings models. This is strongly suggestive that the authors key result isn't about stochastic population dynamics per se, but instead related to

arguing that variance in reproductive success could change through time. In fact, I believe that the result of Kaj and Krone (2003) is substantially more general than the models considered in this manuscript. That being said, I believe that the authors of this manuscript do a much better job of making the implications for evolutionary processes clear than Kaj and Krone, which is important—it's very difficult to understand from Kaj and Krone the conditions under which effective population sizes will be substantially impacted by stochastic population dynamics.

I also find the authors exposition on Paradox 3 to be somewhat strange. First of all, I'm not sure there's a paradox there at all? The authors claim that the lack of dependence of the fixation probability on N_e is a paradox, but this is ultimately not surprising—fixation of a positively selected allele depends mostly on escaping the boundary layer, which doesn't really depend on the population size (see Gillespie's book "The Causes of Molecular Evolution" for great exposition on boundary layer effects). Moreover, the authors *use a Cannings-style argument* to get gain a good approximation of how the fixation probability changes when there is non-Poisson reproduction. So it's not clear that the WFH model is really doing a lot of work here. I suppose they raise the interesting point that the particularly simple form of $p(\text{fix}) = 2s$ is due to the assumption that variance in offspring is equal to 1.

In addition, I raised some concerns about the analysis of empirical results on reproductive variance in my original review, and I don't believe that the authors responded to it at all. I'm not super worried about that analysis, but I think that the authors should probably respond to me.

Overall, I feel like I now have a better understanding of this manuscript. However, I think it still presents its results too strongly: Paradox 1 contains important words of caution that reflect what I am confident is an under appreciated possibility, and Paradox 3 is, as far as I'm concerned, not a paradox at all. I have not addressed Paradox 2 very much because I think that another reviewer had solid and interesting comments on that front and I am leaving it to them. That being said, I do think Paradox 2 actually presents a deep problem in the literature and that the authors' argument may actually represent a path toward a solution.

This manuscript can be a useful contribution to the literature, but as it's presented at the moment, I think most of it is worded too strongly and it continues to not engage appropriately with the literature. Theoretical advances are undoubtedly important, and I think the manuscript presents some interesting things to think about, but ultimately needs to be better situated and several of the claims strongly toned down.

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Reviewer #2 (Public review):

Summary:

This theoretical paper examines genetic drift in scenarios deviating from the standard Wright-Fisher model. The authors discuss Haldane's branching process model, highlighting that the variance in reproductive success equates to genetic drift. By integrating the Wright-Fisher model with the Haldane model, the authors derive theoretical results that resolve paradoxes related to effective population size.

Strengths:

The most significant and compelling result from this paper is perhaps that the probability of fixing a new beneficial mutation is $2s/V(K)$. This is an intriguing and potentially generalizable discovery that could be applied to many different study systems.

The authors also made a lot of effort to connect theory with various real-world examples, such as genetic diversity in sex chromosomes and reproductive variance across different species.

Comments on previous revisions:

The author has addressed some of the concerns in my review, and I think the revised manuscript is more clear. I like the discussion about the caveats of the WFH model.

I hope the authors could also discuss the conditions needed for $V(K)/N_e$ to be a reasonable approximation. It is currently unclear how the framework should be adopted in general.

The idea about estimating male-female $V(K)$ ratios from population genetic data is interesting. Unfortunately, the results fell short. The accuracy of their estimators (derived using approximation $N_e/V(K)$ approximation, and certain choice of theta, and then theta estimated with Watterson's estimator) should be tested with simulated results before applying to real data. The reliability of their estimator and their results from real data are unclear.

Arguments made in this paper sometimes lack precision (perhaps the authors want to emphasize intuition, but it seems more confusing than otherwise). For example: The authors stated that "This independence from N seems intuitively obvious: when an advantageous mutation increases to say, 100 copies in determining a population (depending mainly on s), its fixation would be almost certain, regardless of N ". Assuming large N_e , and with approximation, one could assume the probability of loss is $e^{-(2sn)}$, but the writing about "100 copies" and "almost certain" is very imprecise, in fact, a mutation with $s=0.001$ segregating at 100 copies in a large N_e population is most probably lost. Whereas in a small population, it will be fixed. Yet the following sentence states "regardless of N . This may be a most direct argument against equating genetic drift, certainly no less important than $1/N$ with N , or N_e (which is supposed to be a function of N 's)." I find this new paragraph misleading.

Some of the statements/wordings in this paper still seem too strong to me.

Comments on revisions:

The authors toned down. I am a bit confused because I do not seem to find any point-to-point response to my review.

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Reviewer #3 (Public review):

Summary:

Ruan and colleagues consider a branching process model (in their terminology the "Haldane model") and the most basic Wright-Fisher model. They convincingly show that offspring distributions are usually non-Poissonian (as opposed to what's assumed in the Wright-Fisher model), and can depend on short-term ecological dynamics (e.g., variance in offspring number may be smaller during exponential growth). The authors discuss branching processes and the Wright-Fisher model in the context of 3 "paradoxes" --- 1) how N_e depends on N might depend on population dynamics; 2) how N_e is different on the X chromosome, the Y chromosome, and the autosomes, and these differences do match the expectations base on

simple counts of the number of chromosomes in the populations; 3) how genetic drift interacts with selection. The authors provide some theoretical explanations for the role of variance in the offspring distribution in each of these three paradoxes. They also perform some experiments to directly measure the variance in offspring number, as well as perform some analyses of published data.

Strengths:

- The theoretical results are well-described and easy to follow.
- The analyses of different variances in offspring number (both experimentally and analyzing public data) are convincing that non-Poissonian offspring distributions are the norm.
- The point that this variance can change as the population size (or population dynamics) change is also very interesting and important to keep in mind.
- I enjoyed the Density-Dependent Haldane model. It was a nice example of the decoupling of census size and effective size.
- Equation (10) is a nice result

Comments on revisions:

I appreciate the effort that the authors have put into the revision, but I still find the framing to be a bit confusing – these apparent paradoxes only appear in the most basic version of Wright-Fisher models, and so framing the paper as the solution to these paradoxes overlooks much previous work. Saying that existing work discussing exactly these phenomena is "beyond the scope of this study", without citing or interacting in any way with that work is unscholarly. I agree with the authors that the apparent paradoxes that they consider and interesting, and by thinking about branching processes, the apparent paradoxes appear to be less paradoxical, but without contextualizing this work in the substantial Wright-Fisher literature (e.g., Cannings Exchangeable Models and the work of Möhle) it misrepresents the state of the field and the contributions of this paper.

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Author response:

The following is the authors' response to the previous reviews.

eLife Assessment (divided into 3 parts)

This study presents a useful modification of a standard model of genetic drift by incorporating variance in reproductive success, claiming to address several paradoxes in molecular evolution.

It is crucial to emphasize that our model is NOT a modification of the standard model. The Haldane model, which is generalized here for population regulation, is based on the branching process. The Haldane model and the WF model which is based on population sampling are fundamentally different. We referred to our model as the integrated WF-H model because the results obtained from the WF model over the last 90 years are often (but not always) good approximations for the Haldane model. The analogy would be the comparisons between the Diffusion model and the Coalescence model. Obviously, the results from one model are often good approximations for the other. But it is not right to say that one is a useful modification of the other.

We realize that it is a mistake to call our model the integrated WFH model, thus causing confusions over two entirely different models. Clearly, the word "integrated" did not help. We have now revised the paper by using the more accurate name for the model – the Generalized

Haldane (GH) model. The text explains clearly that the original Haldane model is a special case of the GH model.

Furthermore, we present the paradoxes and resolve them by the GH model. We indeed overreached by claiming that WF models could not resolve them. Whether the WF models have done enough to resolve the paradoxes or at least will be able to resolve them should not be a central point of our study. Here is what we state at the end of this study.:

“We understand that further modifications of the WF models may account for some or all of these paradoxes. However, such modifications have to be biologically feasible and, if possible, intuitively straightforward. Such possible elaborations of WF models are beyond the scope of this study. We are only suggesting that the Haldane model can be extensively generalized to be an alternative approach to genetic drift. The GH model attempts to integrate population genetics and ecology and, thus, can be applied to genetic systems far more complex than those studied before. The companion study is one such example.”

..... However, some of the claimed "paradoxes" seem to be overstatements, as previous literature has pointed out the limitations of the standard model and proposed more advanced models to address those limitations....

As stated in the last paragraph of the paper, it is outside of the scope of our study to comment on whether the earlier WF models can resolve these paradoxes. So, all such statements have been removed or at least drastically toned down in the formal presentation. That said, editors and reviewers may ask whether we are re-inventing the wheels. The answers are as follows:

First, two entirely different models reaching the same conclusion are NOT the re-invention of wheels. The coalescence theory does not merely rediscover the results obtained by the diffusion models. The process of obtaining the results is itself a new invention. This would lead to the next question: is the new process more rigorous and more efficient? I think the Haldane model is indeed more efficient in comparisons with the very complex modifications of the WF models.

Second, we are not sure that the paradoxes have been resolved, or even can be resolved. Note that these skepticisms have been purged from the formal presentation. Therefore, I am presenting the arguments outside of the paper for a purely intellectual discourse. Below, please allow us to address the assertions that the WF models can resolve the paradoxes.

The first paradox is that the drift strength in relation to N is often opposite of the WF model predictions. Since the WF models (standard or modified) do not generate N from within the model, how can it resolve the paradox? In contrast, the Generalized Haldane model generates N within the model. It is the regulation of N near the carrying capacity that creates the paradox – When N increases, drift also increases.

The second paradox that the same locus experiences different drifts in males and females is accepted by the reviewers. Nevertheless, we would like to point out that this second paradox echoed the first one as newly stated in the Discussion section “The second paradox of sex-dependent drift is about different $V(K)$ ’s between sexes (generally $V_m > V_f$) but the same $E(K)$ between them. In the conventional models of sampling, it is not clear what sort of biological sampling scheme could yield $V(K) \neq E(K)$, let alone two separate $V(K)$ ’s with one single $E(K)$. Mathematically, given separate K distributions for males and females, it is unlikely that $E(K)$ for the whole population could be 1, hence, the population would either explode in size or decline to zero. In short, N regulation has to be built into the genetic drift model as the GH model does to avoid this paradox.”

The third paradox stems from the fact that drift is operating even for genes under selection. But then the drift strength, $2s/V(K)$ for an advantage of s , is independent of N or N_e . Since the

determinant of drift strength in the WF model is ALWAYS Ne, how is Paradox 3 not a paradox for the WF model?

The 4th paradox about multi-copy gene systems is the subject of the companion paper (Wang et al.). Note that the WF model cannot handle systems of evolution that experience totally different sorts of drift within vs. between hosts (viruses, rDNAs etc). This paradox can be understood by the GH model and will be addressed in the next paper.

While the modified model presented in this paper yields some intriguing theoretical predictions, the analysis and simulations presented are incomplete to support the authors' strong claims, and it is unclear how much the model helps explain empirical observations.

The objections appear to be that our claims of “paradox resolution” being too strong. We interpret this objection is based on the view (which we agree) that these paradoxes are intrinsically difficult to resolve by the WF models. Since our model has been perceived to be a modified WF model, the claim of resolution is clearly too strong. However, the GH model is conceptually and operationally entirely different from the WF models as we have emphasized above. In case our reading of the editorial comments is incorrect, would it be possible for some clarifications on the nature of “incomplete support”? We would be grateful for the help.

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